



Final Report

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Mutual Fund Analytics Platform

*Prepared
By*

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Technologies:

Python, SQL, Power BI, Jupyter Notebook

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1. EXECUTIVE SUMMARY

This project developed an end-to-end Mutual Fund Analytics Platform for Bluestock Fintech. The solution integrates multiple mutual fund datasets, performs ETL processing, stores data in a structured analytical model, computes advanced risk and performance metrics, and delivers insights through an interactive Power BI dashboard.

The platform analyzes mutual fund performance, investor behavior, SIP trends, fund risk characteristics, and benchmark comparisons. Advanced analytics including Value at Risk (VaR), Conditional Value at Risk (CVaR), cohort analysis, SIP continuity assessment, and fund recommendation logic were implemented.

The final solution enables investors and analysts to make data-driven decisions using a centralized analytics platform.

2. DATA SOURCES

Overview

The Mutual Fund Analytics Platform was developed using multiple datasets covering mutual fund performance, assets under management, benchmark indices, investor transactions, SIP activity, portfolio holdings, and industry-level metrics. These datasets were integrated to provide a comprehensive view of the Indian mutual fund industry from both investor and fund management perspectives.

Data Sources Used

1. Fund Master Dataset

Contains static information about mutual fund schemes including scheme name, fund house, category, plan type, launch date, benchmark, expense ratio, and minimum investment requirements.

Key Attributes:

- AMFI Code
- Scheme Name
- Fund House
- Category and Sub-Category
- Plan Type
- Launch Date
- Benchmark Index
- Expense Ratio

2. NAV Dataset

Contains daily Net Asset Value (NAV) observations for mutual fund schemes between 2022 and 2025.

Key Attributes:

- AMFI Code
- Date
- NAV
- Daily Return

Purpose:

- NAV trend analysis
- Return calculations
- Risk analytics
- VaR and CVaR computation

3. Performance Dataset

Contains fund performance and risk-adjusted return metrics.

Key Attributes:

- 1-Year Return
- 3-Year Return
- 5-Year Return
- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio
- Standard Deviation
- Maximum Drawdown

Purpose:

- Performance comparison
- Risk-return analysis
- Recommendation engine

4. AUM Dataset

Tracks Assets Under Management across mutual fund houses.

Key Attributes:

- Fund House
- AUM (₹ Crore)

Purpose:

- Industry size analysis
- Fund house comparison
- Market share evaluation

5. Category Inflows Dataset

Contains monthly inflows and outflows across mutual fund categories.

Key Attributes:

- Month
- Category
- Net Inflow (₹ Crore)

Purpose:

- Category preference analysis
- Investor sentiment tracking
- Sector concentration analysis

6. SIP Industry Dataset

Provides monthly SIP statistics for the mutual fund industry.

Key Attributes:

- SIP Inflow

- Active SIP Accounts
- SIP AUM
- Year-over-Year Growth

Purpose:

- SIP trend analysis
- Investor participation assessment

7. Investor Transactions Dataset

Contains individual investor transaction records.

Key Attributes:

- Investor ID
- Transaction Date
- Transaction Type
- Investment Amount
- Age Group
- Income Level
- Payment Mode
- Geographic Information

Purpose:

- Cohort analysis
- SIP continuity analysis
- Investor behavior analytics

8. Portfolio Holdings Dataset

Contains portfolio allocation across sectors and asset classes.

Purpose:

- Portfolio diversification analysis
- Concentration measurement
- Sector allocation review

9. Benchmark Dataset

Contains benchmark index values used for performance comparison.

Purpose:

- Benchmark tracking
- Relative performance measurement
- Market trend analysis

Data Quality and Preparation

All datasets underwent a structured ETL pipeline consisting of extraction, validation, cleaning, transformation, and loading stages. Missing values, duplicate records, formatting inconsistencies, and datatype mismatches were identified and resolved before analytical processing.

The final analytical datasets were stored in a structured format and used for SQL analysis, Python-based analytics, and Power BI dashboard development.

3. ETL PIPELINE AND DATA ENGINEERING

Overview

An end-to-end ETL (Extract, Transform, Load) pipeline was developed to collect, clean, transform, and prepare mutual fund datasets for analytical processing. The objective of the pipeline was to ensure data consistency, reliability, and usability for downstream SQL analysis, Python-based analytics, and Power BI dashboard development.

The ETL process was implemented using Python, Pandas, and PostgreSQL.

ETL Architecture

The ETL workflow consisted of three major stages:

Extract

Data was collected from multiple structured CSV datasets containing:

- Fund Master Information
- Daily NAV Records
- Mutual Fund Performance Metrics
- Assets Under Management (AUM)
- Category Inflows
- SIP Industry Statistics
- Investor Transactions
- Portfolio Holdings
- Benchmark Indices

The extracted datasets were loaded into Pandas DataFrames for preprocessing and validation.

Transform

The transformation stage focused on improving data quality and creating analysis-ready datasets.

The following transformations were performed:

Data Cleaning

- Removed duplicate records

- Standardized column names
- Converted date columns into datetime format
- Corrected datatype inconsistencies
- Trimmed unwanted spaces and formatting issues

Missing Value Handling

- Identified missing values across datasets
- Applied appropriate imputation techniques where required
- Removed records with critical missing information

Feature Engineering

Several derived metrics were generated, including:

- Daily Returns from NAV data
- Rolling Sharpe Ratios
- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- SIP Continuity Metrics
- Cohort Analysis Metrics
- Risk Classification Indicators

Validation Rules

Data validation checks included:

- Duplicate detection
- Negative value checks
- Expense ratio validation
- Sharpe ratio validation
- Category consistency checks
- Date range verification

Load

After cleaning and transformation, the datasets were loaded into PostgreSQL.

The database schema was designed to support analytical querying and dashboard reporting.

Processed datasets were also exported as cleaned CSV files for reproducibility and future analysis.

Database Design

The PostgreSQL database consisted of multiple analytical tables:

Table Name	Purpose
fund_master	Scheme metadata and fund details
nav	Daily NAV history
performance	Fund performance metrics
aum	Assets under Management data
category_inflows	Monthly category flows
sip_industry	SIP statistics
transactions	Investor transaction records
portfolio_holdings	Portfolio allocation data
benchmark_indices	Market benchmark information

The relational structure enabled efficient joins and analytical queries across multiple business domains.

Data Processing Workflow

1. Raw CSV datasets were extracted.
2. Data quality checks were performed.
3. Missing and duplicate values were handled.
4. New analytical features were created.
5. Cleaned datasets were stored in PostgreSQL.
6. Processed datasets were exported for analytics.
7. SQL, Python, and Power BI consumed the final analytical layer.

Outcome

The ETL pipeline transformed raw mutual fund data into a structured analytical environment suitable for business intelligence reporting, statistical analysis, risk modeling, and investor behavior analytics. The resulting datasets served as the foundation for all subsequent analyses and dashboard visualizations developed during the project.

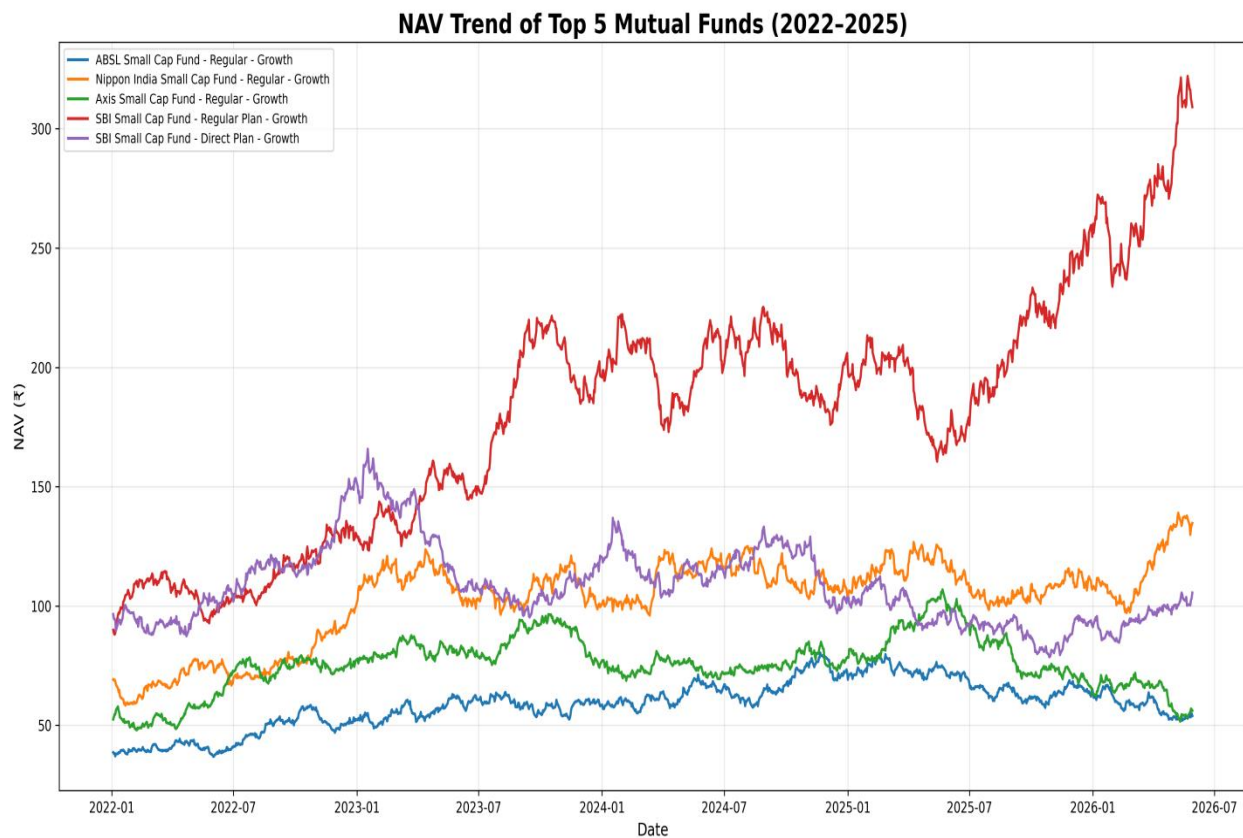
4. EXPLORATORY DATA ANALYSIS (EDA) AND KEY FINDINGS

Overview

Exploratory Data Analysis (EDA) was conducted to understand trends, patterns, and relationships within the mutual fund ecosystem. The analysis focused on fund growth, industry expansion, investor participation, category preferences, and benchmark comparisons.

The insights generated during this phase guided the development of analytical models and dashboard visualizations.

4.1 NAV Trend Analysis



Observation

The NAV trends of the top-performing mutual fund schemes were analyzed over the period 2022–2025.

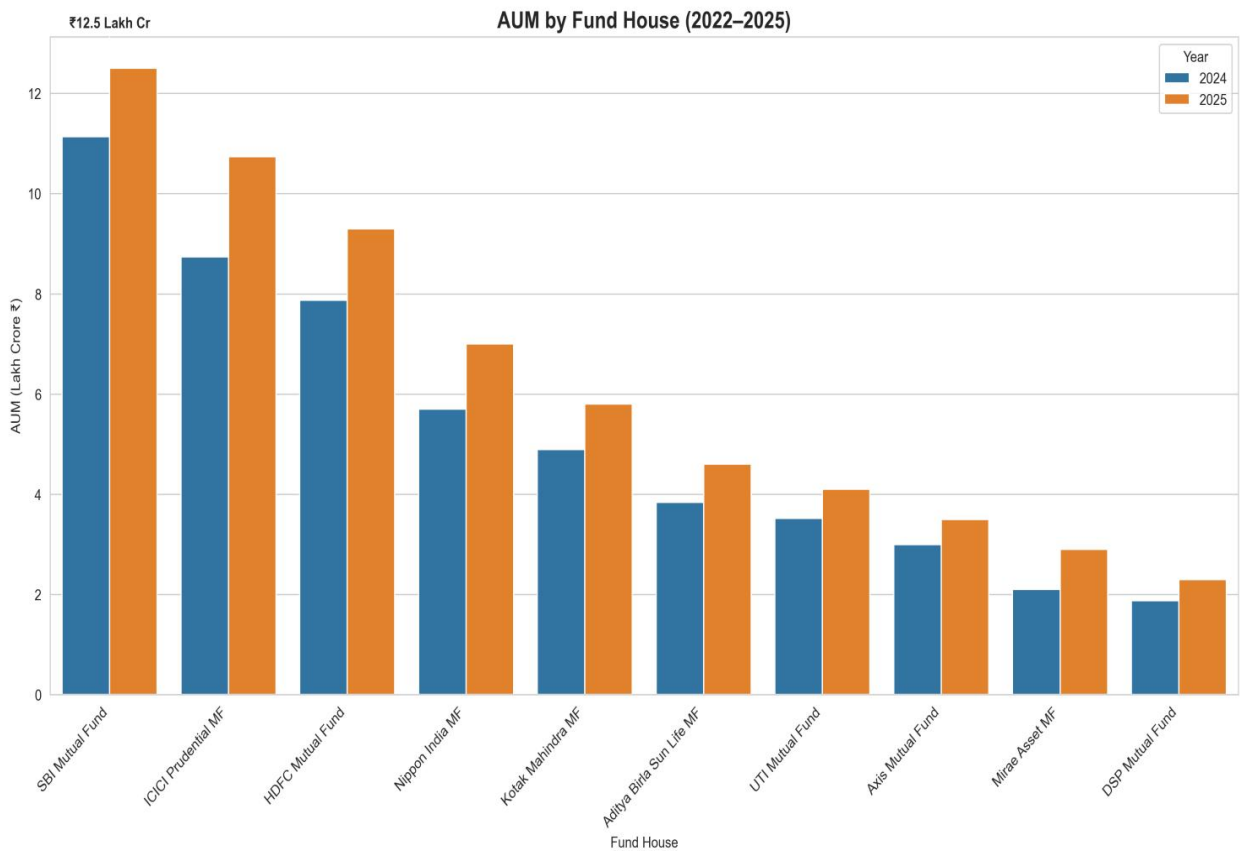
Key Findings

- Most top-performing funds exhibited a consistent upward trend over the analysis period.
- Small-cap funds demonstrated higher growth potential but also showed greater volatility.
- Certain schemes significantly outperformed their peers, indicating superior long-term wealth creation.
- Market corrections were visible across all schemes, reflecting broader market movements.

Business Insight

Investors seeking higher returns may benefit from exposure to well-performing equity funds, although such funds typically involve higher risk and volatility.

4.2 Assets Under Management (AUM) Analysis



Observation

AUM distribution was analyzed across leading mutual fund houses for 2024 and 2025.

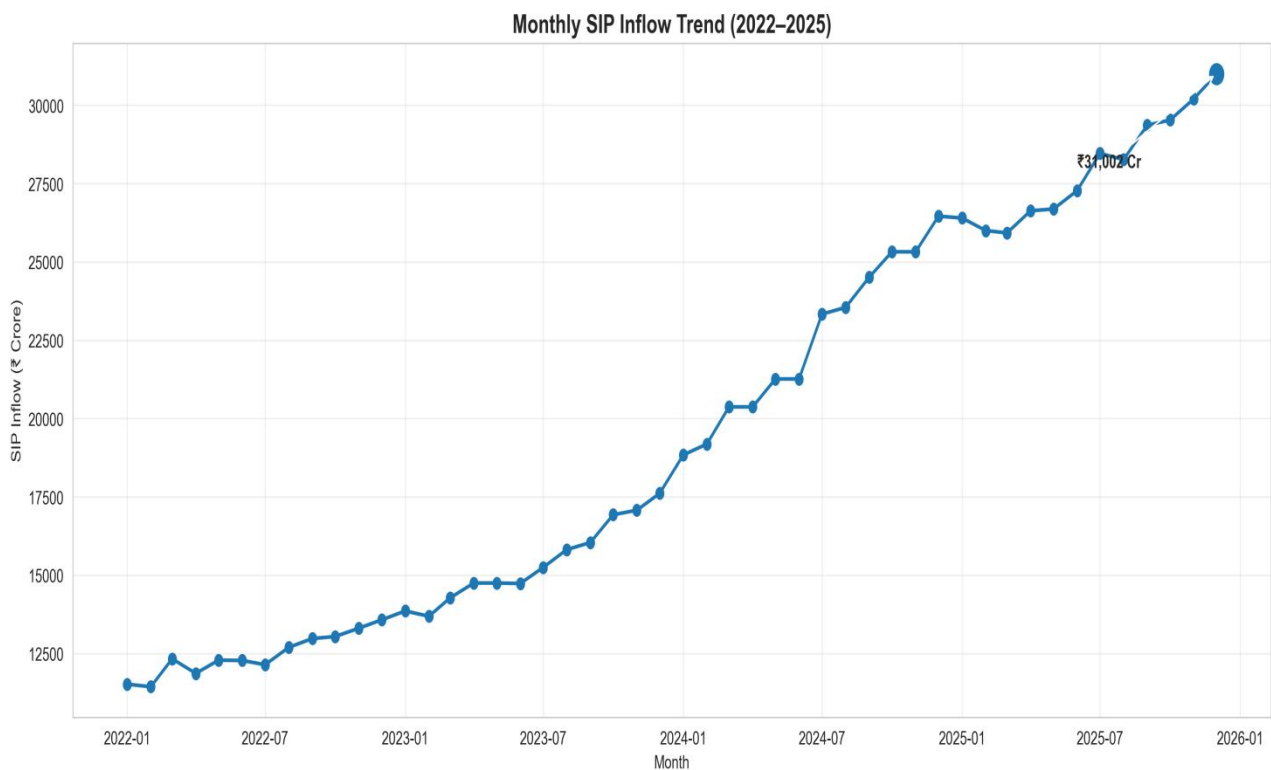
Key Findings

- SBI Mutual Fund maintained the highest AUM among all fund houses.
- ICICI Prudential Mutual Fund and HDFC Mutual Fund emerged as strong competitors with substantial asset bases.
- Most fund houses recorded AUM growth from 2024 to 2025.
- Industry assets demonstrated broad-based expansion rather than concentration in a single institution.

Business Insight

Growth in AUM indicates increasing investor confidence and industry expansion. Larger fund houses continue to dominate due to strong brand recognition and distribution networks.

4.3 SIP Industry Growth Analysis



Observation

Monthly SIP inflows were examined from 2022 to 2025.

Key Findings

- SIP inflows increased steadily throughout the period.
- Monthly inflows grew from approximately ₹11,500 crore in early 2022 to over ₹31,000 crore by the end of 2025.
- Temporary slowdowns were followed by strong recoveries.
- The long-term trend remained strongly positive.

Business Insight

The consistent growth in SIP participation highlights increasing retail investor adoption of disciplined long-term investing practices.

4.4 Category-wise Inflow Analysis



Observation

Net inflows across mutual fund categories were analyzed using a heatmap.

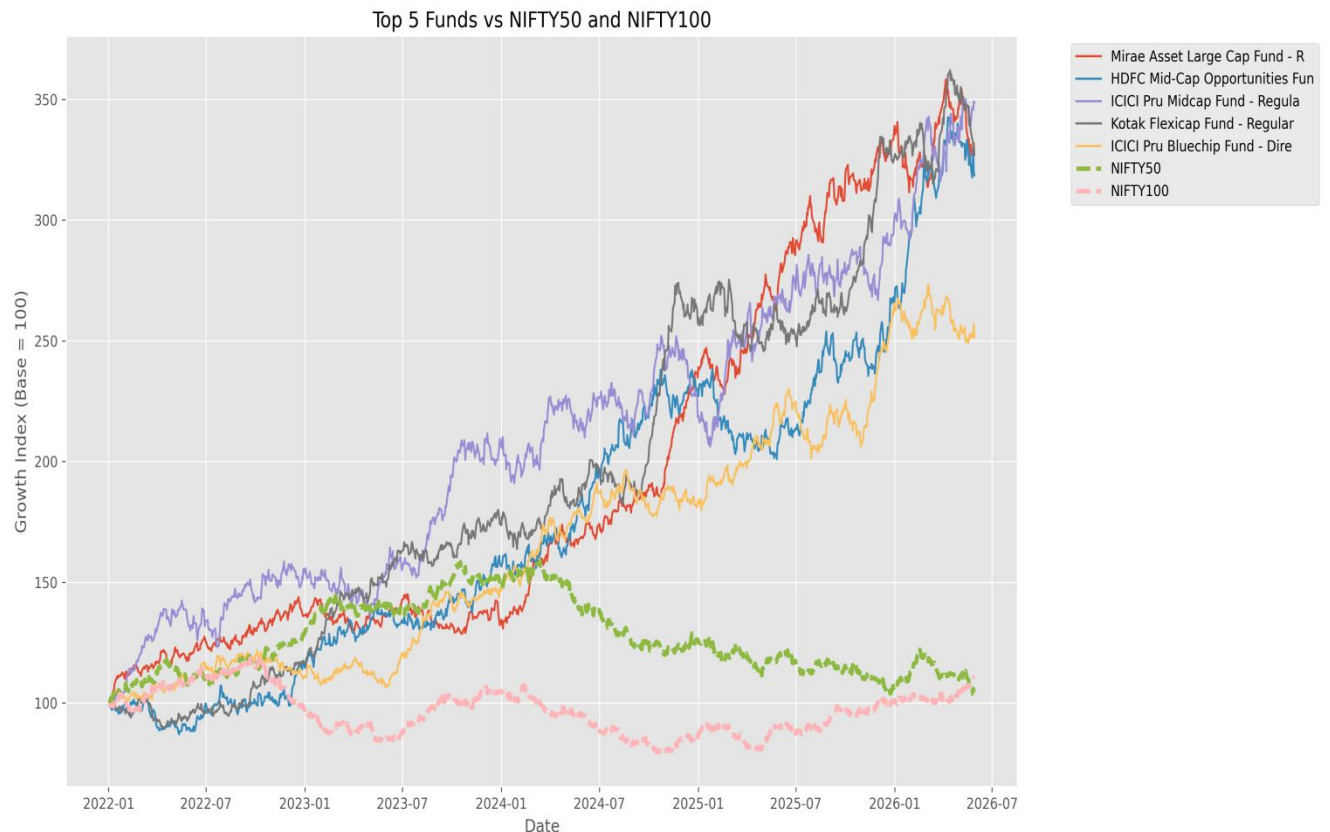
Key Findings

- Liquid Funds consistently attracted the highest inflows across most months.
- Sectoral/Thematic Funds received strong investor interest during favorable market conditions.
- Flexi Cap and Large & Mid Cap categories maintained stable inflow levels.
- ELSS and Gilt Funds attracted comparatively lower inflows.

Business Insight

Investor preference remains concentrated in categories offering liquidity, diversification, and tactical exposure to market themes.

4.5 Benchmark Comparison Analysis



Observation

Top-performing mutual funds were compared against benchmark indices including NIFTY50 and NIFTY100.

Key Findings

- Several mutual funds outperformed benchmark indices over the analysis period.
- Active fund management generated excess returns in selected schemes.
- Benchmark indices displayed lower growth compared with leading funds.
- Performance differences became increasingly visible over longer investment horizons.

Business Insight

Fund selection remains critical, as high-quality actively managed funds can significantly outperform passive benchmark returns over time.

EDA Summary

The exploratory analysis revealed strong industry growth, increasing SIP participation, rising AUM across fund houses, and meaningful performance differentiation among mutual fund categories and schemes. These findings established the foundation for advanced risk analytics, investor behavior analysis, recommendation systems, and dashboard development carried out in later stages of the project.

5. ADVANCED ANALYTICS AND RISK ANALYSIS

Overview

Beyond traditional reporting and visualization, advanced analytical techniques were applied to evaluate mutual fund risk, investor behavior, portfolio concentration, and recommendation strategies. These models provide deeper insights into fund performance and investor decision-making.

The analytical framework includes risk assessment, cohort analysis, SIP continuity monitoring, fund recommendation logic, and concentration measurement.

5.1 Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Objective

To estimate potential downside risk associated with mutual fund investments under adverse market conditions.

Methodology

Daily returns were calculated using historical NAV data. The 95% confidence level was used to estimate:

- Value at Risk (VaR)
- Conditional Value at Risk (CVaR)

VaR estimates the maximum expected loss under normal market conditions, while CVaR measures the average loss beyond the VaR threshold.

Key Findings

- Small-cap and mid-cap funds exhibited higher downside risk compared with large-cap funds.
- CVaR values consistently exceeded VaR values, indicating tail-risk exposure.
- Risk profiles varied significantly across fund categories.

Business Insight

VaR and CVaR metrics provide investors with a quantitative understanding of potential losses and assist in risk-adjusted investment decisions.

5.2 Rolling Sharpe Ratio Analysis

Objective

To evaluate risk-adjusted performance over time.

Methodology

Rolling Sharpe Ratios were calculated using historical return series.

The metric compares excess returns against volatility and enables continuous monitoring of fund efficiency.

Key Findings

- High-performing funds maintained consistently positive Sharpe Ratios.
- Market volatility periods resulted in temporary Sharpe Ratio declines.
- Funds with stable Sharpe Ratios generally demonstrated superior long-term performance.

Business Insight

Rolling Sharpe analysis enables investors to assess whether higher returns are being generated efficiently relative to risk.

5.3 Investor Cohort Analysis

Objective

To analyze investor participation patterns across different investment cohorts.

Methodology

Investors were grouped according to their first recorded transaction year and tracked over time.

Key Findings

- The 2024 investor cohort represented the majority of active investors.
- New investor acquisition remained positive throughout the analysis period.

- Larger cohorts generated significantly higher cumulative investments.

Business Insight

Cohort analysis helps identify growth trends, retention patterns, and the effectiveness of investor acquisition strategies.

5.4 SIP Continuity Analysis

Objective

To identify investors at risk of discontinuing systematic investment plans (SIPs).

Methodology

Transaction frequency and average investment gaps were analyzed.

Investors were classified into:

- Healthy
- At Risk

based on SIP continuity behavior.

Key Findings

- A substantial proportion of investors exhibited irregular SIP patterns.
- Larger transaction gaps were associated with increased discontinuation risk.
- SIP participation remained concentrated among a subset of highly active investors.

Business Insight

Early identification of at-risk investors enables targeted engagement strategies and improves long-term customer retention.

5.5 Fund Recommendation Engine

Objective

To recommend suitable mutual funds based on investor risk preferences.

Methodology

A rule-based recommendation system was developed using:

- Fund Category
- Risk Profile
- Sharpe Ratio Ranking

Investors were categorized into:

- Low Risk
- Moderate Risk
- High Risk

Top-ranked funds were selected using risk-adjusted performance metrics.

Key Findings

- Different investor profiles received distinct recommendations.
- Sharpe Ratio proved effective for ranking risk-adjusted performance.
- Recommendations aligned with category-specific risk characteristics.

Business Insight

The recommendation engine demonstrates how analytical models can support personalized investment decision-making.

5.6 Sector Concentration Analysis (HHI)

Objective

To measure concentration risk across mutual fund categories.

Methodology

The Herfindahl-Hirschman Index (HHI) was calculated using category-level inflow market shares.

The index measures concentration by summing the squared market shares of individual categories.

Key Findings

- Certain categories attracted a disproportionately large share of industry inflows.
- Liquid Funds represented a significant portion of total inflows.
- Industry allocation was moderately concentrated.

Business Insight

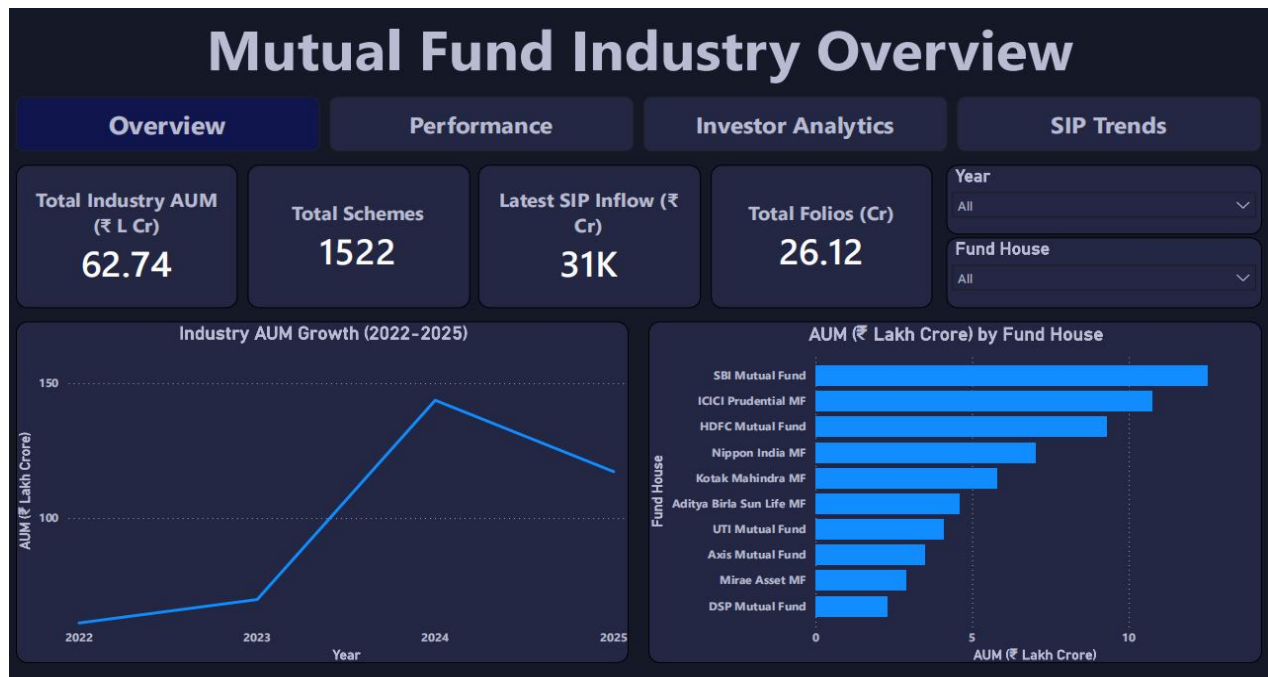
Concentration metrics help identify dependency on specific categories and highlight diversification opportunities.

Advanced Analytics Summary

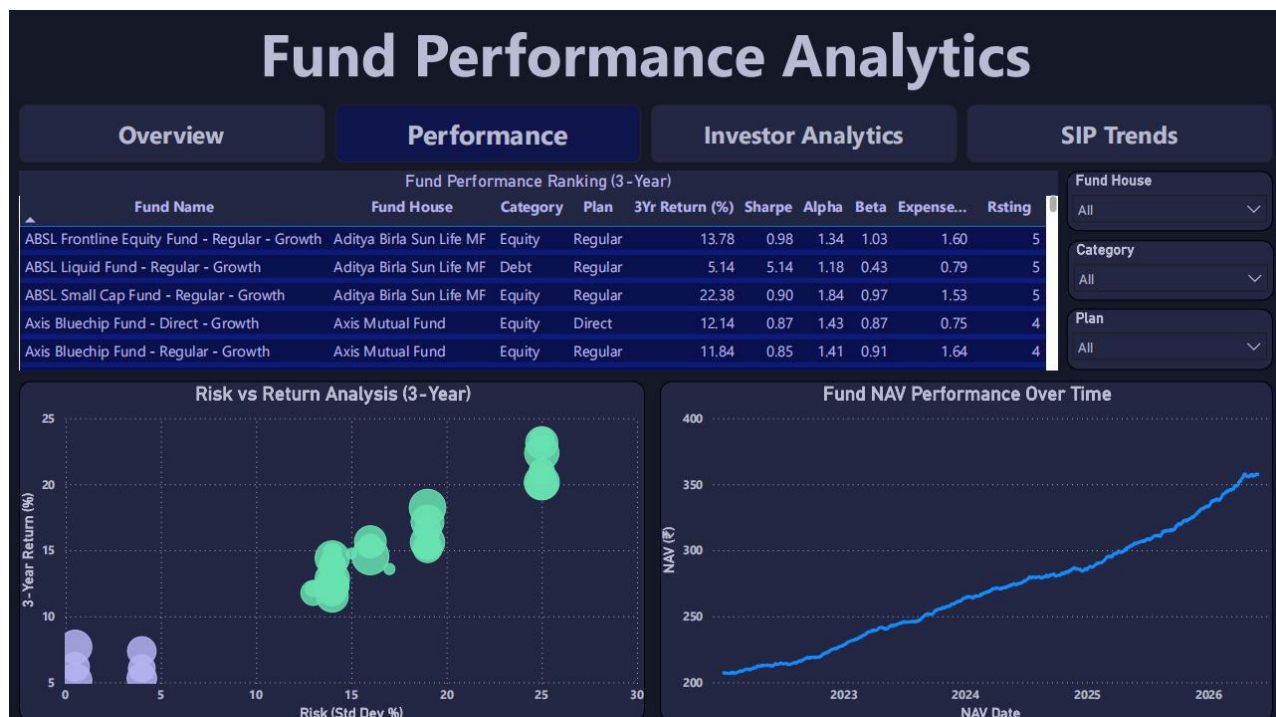
The advanced analytical framework provided insights beyond descriptive reporting by quantifying risk exposure, investor behavior, portfolio concentration, and fund suitability. These models enhance decision-making capabilities for investors, analysts, and financial institutions and demonstrate the practical application of data science within the mutual fund industry.

6. POWER BI DASHBOARD DEVELOPEMENT

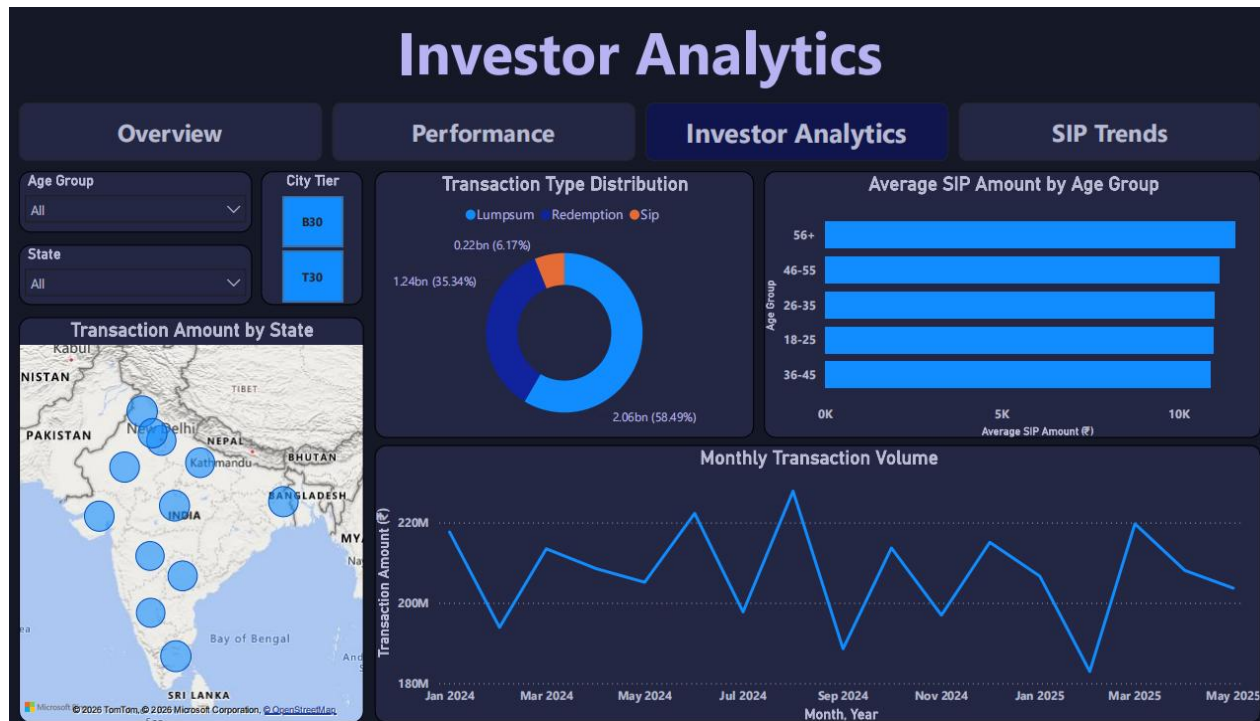
6.1 Overview Dashboard



6.2 Fund Performance Analytics Dashboard



6.3 Investor Analytics Dashboard



6.4 SIP & Market Trends Dashboard



7. BUSINESS RECOMMENDATIONS

Overview

Based on the exploratory analysis, performance evaluation, investor behavior assessment, and dashboard insights, several strategic recommendations can be made for mutual fund houses, investment advisors, and investors.

7.1 Promote SIP-Based Investing

The analysis revealed a strong and consistent increase in monthly SIP inflows throughout the study period. This indicates growing investor confidence in systematic investment plans as a long-term wealth creation strategy.

Recommendation

- Encourage long-term SIP participation through investor education programs.
- Introduce goal-based SIP products tailored to different investor segments.
- Use digital engagement initiatives to improve SIP retention.

Expected Benefit

Higher investor retention, increased recurring inflows, and stronger long-term asset growth.

7.2 Strengthen Investor Retention Programs

SIP continuity analysis identified a segment of investors exhibiting irregular investment behavior and higher discontinuation risk.

Recommendation

- Develop early-warning systems to identify at-risk investors.
- Implement targeted communication campaigns and investment reminders.
- Offer personalized portfolio reviews to inactive investors.

Expected Benefit

Reduced SIP discontinuation rates and improved customer lifetime value.

7.3 Expand High-Performing Fund Categories

Performance analysis demonstrated that certain equity-oriented categories consistently delivered superior long-term returns.

Recommendation

- Increase visibility of high-performing categories through marketing initiatives.
- Educate investors regarding risk-adjusted return opportunities.
- Develop new products aligned with successful category strategies.

Expected Benefit

Higher investor participation and improved product adoption.

7.4 Improve Risk Awareness and Transparency

Advanced risk analytics highlighted meaningful differences in downside risk across mutual fund categories.

Recommendation

- Include VaR, CVaR, and risk-adjusted metrics within investor reporting.
- Improve communication regarding volatility and investment risk.
- Provide category-specific risk disclosures.

Expected Benefit

Better-informed investment decisions and improved investor confidence.

7.5 Leverage Data-Driven Fund Recommendations

The recommendation engine demonstrated the potential of analytics-driven investment guidance.

Recommendation

- Integrate recommendation models into investor-facing applications.
- Personalize investment suggestions based on risk tolerance and objectives.
- Continuously refine recommendation logic using investor behavior data.

Expected Benefit

Enhanced user experience and improved investment suitability.

7.6 Monitor Category Concentration Risk

Sector concentration analysis showed that investor inflows are concentrated within a limited number of categories.

Recommendation

- Encourage portfolio diversification across multiple categories.
- Monitor concentration levels periodically using HHI metrics.
- Promote balanced asset allocation strategies.

Expected Benefit

Reduced concentration risk and improved portfolio stability.

Business Recommendation Summary

The findings suggest that mutual fund organizations can significantly improve investor engagement, retention, and portfolio outcomes by combining advanced analytics, personalized recommendations, risk transparency, and systematic investment strategies. Data-driven decision-making will play an increasingly important role in supporting long-term industry growth and investor success.

8. PROJECT LIMITATIONS

Overview

While the Mutual Fund Analytics Platform provides valuable insights into fund performance, investor behavior, and industry trends, certain limitations should be acknowledged when interpreting the results.

8.1 Limited Dataset Coverage

The analysis was conducted using a selected set of mutual fund schemes and industry datasets available during the project period.

Impact

- Results may not represent the entire mutual fund universe.
- Additional schemes and historical records could provide broader market coverage.

8.2 Historical Data Dependency

Most analytical models rely on historical performance and transaction data.

Impact

- Past performance does not guarantee future returns.
- Market conditions may change significantly over time.

8.3 Simplified Recommendation Engine

The recommendation system was implemented using rule-based logic and risk categorization.

Impact

- Recommendations do not account for individual investor goals, investment horizons, or financial situations.
- More advanced machine learning approaches could improve personalization.

8.4 Limited Market Factors

The analysis primarily focused on mutual fund and investor datasets.

Impact

- Macroeconomic indicators, interest rates, inflation, and global market events were not incorporated.
- External market influences may affect actual investment outcomes.

8.5 Assumptions in Risk Models

Risk metrics such as VaR and CVaR were calculated using historical return distributions.

Impact

- Extreme market events may not be fully captured.
- Risk estimates should be interpreted as approximations rather than guarantees.

8.6 Dashboard Scope

The Power BI dashboard was designed to demonstrate key analytical capabilities within the project timeline.

Impact

- Additional drill-through functionality, forecasting models, and real-time integrations could further enhance the platform.

Limitation Summary

Despite these limitations, the project successfully demonstrates the application of data engineering, analytics, risk modeling, and business intelligence techniques within the mutual fund domain. The platform provides a strong foundation for future enhancements and real-world deployment.

9. CONCLUSION

The Mutual Fund Analytics Platform successfully delivered an end-to-end analytical solution for understanding mutual fund performance, investor behavior, industry growth, and market trends.

The project integrated multiple datasets through a structured ETL pipeline, transformed raw data into analysis-ready formats, and stored the information in a relational database for efficient querying and reporting. Exploratory analysis revealed meaningful trends in NAV growth, SIP participation, category preferences, and AUM expansion across leading fund houses.

Advanced analytical techniques including Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio analysis, Cohort Analysis, SIP Continuity Analysis, Fund Recommendation Systems, and Sector Concentration Measurement provided deeper insights beyond traditional reporting.

The Power BI dashboard consolidated these findings into an interactive business intelligence platform consisting of four dedicated analytical views:

- Industry Overview
- Fund Performance Analytics
- Investor Analytics
- SIP & Market Trends

The project demonstrates the practical application of Python, SQL, PostgreSQL, data analytics, and business intelligence tools in solving real-world financial analytics problems.

Overall, the platform enables data-driven decision-making for investors, analysts, and financial institutions while establishing a scalable foundation for future enhancements such as machine learning-based recommendations, predictive analytics, and real-time market integration.